

SLFM: Semi-Supervised Local Community Detection Based on Hyperbolic Flow Matching

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Abstract

Community detection is a longstanding topic in graph and Web algorithms, and **semi-supervised local community detection**, identifying the community to which the given user belongs, garners increasing research attention in recent years. While achieving encouraging results, existing solutions often encounter accumulated errors due to the weak supervision in the community expansion process, and are undermined by the initial seed sensitivity that a suboptimal or boundary seed node can easily misguide community generation. To fill these gaps, we propose **a fresh generative perspective on hyperbolic space**, which recasts this problem as the seed-conditioned sequence generation, and reformulates community generation as a continuous transport of probability distributions in the manifold measure space. In this paper, we present a novel Semi-supervised Local community detection framework based on hyperbolic Flow Matching (**SLFM**). Specifically, it leverages a geometric-aware Seed Selector that refines initial seeds with hyperbolic angular and radial priors, and a Hyperbolic Flow Transporter that learns a vector field to map a source distribution to a target community distribution, generating a robust set of anchors. Finally, a Community Expander is introduced to utilize these anchors as surrogate supervision to effectively recover the full community. Experimental results on four real-world datasets demonstrate that SLFM significantly outperforms existing methods in both local and global semi-supervised settings.

CCS Concepts

• **Information systems** → **Clustering**; • **Mathematics of computing** → **Graph algorithms**.

Keywords

Semi-supervised local community detection; Flow matching; Hyperbolic space

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1 INTRODUCTION

Networks provide a common model for complex relationships across social, information, biological, and scientific domains, where nodes denote entities and edges capture interactions [11, 13, 44]. A crucial step to understand a network is identifying and analyzing closely related subgraphs, which are named community detection [34, 40]. Prior information, such as node attributes, node labels [22], and labeled community [43], can improve community detection. To utilize prior information, some semi-supervised community detection methods have been devised, such as SEAL [43] and CLARE [40].

In some cases, users are only interested in the community to which a given node belongs. To mine a community for a given node, CommunityAF [6] learns a general community structure from labeled communities and uses an autoregressive flow generative model to generate the community containing the query node. SLSS [27] uses a heuristic algorithm that leverages partially labeled communities to guide the generation of a local community to which a given node belongs. SLRL [28] formulates the task as a local sequential decision problem around the query node and refines it via reinforcement learning.

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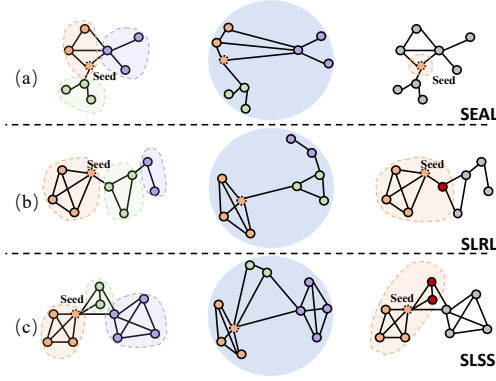


Figure 1: An illustration of seed sensitivity, showing how a suboptimal seed can lead to inaccurate community detection in existing methods. (a) SEAL: Under distribution shift, the seed selector downranks rare structures, so generation never starts for the true community. (b) SLRL: With local-gain rewards, boundary seeds follow external gain paths and stop early. (c) SLSS: Boundary seeds are pulled by external high-similarity cues, causing over-expansion across community boundaries.

Indeed, semi-supervised local community detection can be formulated as seed-conditioned sequence generation [43]. Because real networks are structurally complex, labeled communities seldom cover the full variety of structures. The order of community formation, which would provide strong supervision, is unobserved. In the absence of such supervision, seed expansion reduces to local decisions, and early mistakes propagate and compound over steps [20]. As a result, existing methods often rely on labeled community structures and seed nodes. As shown in Figure 1, due to seed sensitivity, those seed methods fail to recover the true community from a low-quality seed.

Motivated by the demonstrated effectiveness of Flow Matching in graph generation tasks [9, 32, 35, 37, 38, 42], we can construct, in the space of probability measures, a transport path from a seed distribution to a community distribution as a reference trajectory and use it as surrogate supervision to guide the generation from seeds to communities. However, applying Flow Matching to semi-supervised local community detection raises two challenges. (i), how to choose the seed distribution so that, when the seed lies on a community boundary or is a leaf node, the resulting surrogate supervision remains representative and does not misguide generation. (ii), how to represent graph nodes as probabilistic objects that support geometrically consistent transport in continuous time.

To address those issues, we propose a novel method SLFM (Semi-Supervised Local Community Detection Based on Hyperbolic Flow Matching), as shown in Figure 2. Inspired by geometric priors in hyperbolic space, where hyperbolic embeddings arrange communities into angular sectors, while radial coordinates primarily reflect centrality, we address the first challenge with a Seed Selector that relocates the seed within its 1-ego network by computing Angular Consistency and Radial Conformity from the embeddings, together with an augmented Forman–Ricci curvature from graph topology, so that the topological cue complements the hyperbolic priors. For

the second challenge, we embed nodes in hyperbolic space and cast local neighborhoods as Riemannian normal distribution, enabling geodesically faithful, continuous-time transport. In addition, we design a Hyperbolic Flow Transporter, which, together with a graph-structured regularization, learns a time-dependent vector field that transports the source toward the target distribution. Then, it generates an anchor set that captures the surrogate supervision from seeds to communities. In order to establish a mapping between anchor points and communities, we further propose a Community Expander, which aims to recover the community containing the seed by ranking frontier nodes using a hyperbolic kernel coverage gain with respect to the anchor set. It greedily adds the frontier node with the largest positive marginal gain and stops when no further positive gain remains.

We summarize the contributions of this work as follows:

- We reconsider semi-supervised local community detection from a fresh generative perspective on hyperbolic space, and introduce flow matching to this problem for the first time, to the best of our knowledge.
- We design a novel SLFM framework where a geometric prior-based seed selector and a Hyperbolic Flow Transporter to select seeds and learn surrogate supervision, respectively. A surrogate supervision-based community expander is designed to find the community the seed belongs to.
- Experiments are conducted on four real-world datasets. Experimental results show that SLFM achieves better performance than comparison methods.

2 PRELIMINARIES

2.1 Problem formulation

Given a graph $G = (\mathcal{V}, \mathcal{E}, \mathbf{X})$, where $\mathcal{V}$ is the set of nodes, $\mathcal{E}$ is the set of edges, and $\mathbf{X} \in \mathbb{R}^{|\mathcal{V}|}$ is the feature matrix with $\mathbf{x}(u) \in \mathbb{R}$ being the feature vector of node u . With m labeled communities $\mathcal{C}^m = \{\mathcal{C}^{(1)}, \dots, \mathcal{C}^{(m)}\}$ as known community set. For a given seed node u_0 , the goal is to detect the community $\mathcal{C}_{u_0}$ to which u_0 belongs using known community set $\mathcal{C}^m$ [27]. The important notations used are listed in Table 5.

2.2 Flow matching on the manifold

Flow matching is a simulation-free method for learning probability flows, which builds a probability path $(p_t)_{0 \leq t \leq 1}$, from a known source distribution p_0 to the data target distribution p_1 . Recently, flow matching has been extended to general Riemannian manifold.

On a manifold $\mathcal{M}$, a time-dependent vector field v_t induces a probability flow, denoted $\Phi_t(\cdot) : [0, 1] \times \mathcal{M} \rightarrow \mathcal{M}$. The flow Φ_t is defined as the solution to the Ordinary Differential equation:

$$\frac{d}{dt} \Phi_t(x) = v_t(\Phi_t(x)), \quad \Phi_0(x) = x_0, \quad (1)$$

where $v_t(x)$ is a vector in the tangent space $\mathcal{T}_x \mathcal{M}$ at point x . In practice, the vector field v_t that generates the probability path p_t is intractable, as it does not have access to the closed-form of v_t . To address this, flow matching decomposes the target probability p_t as a mixture of tractable conditional probability paths, $p_t(x|x_1)$. Each conditional path satisfies $p_0(x|x_1) = p_0(x)$ and $p_1(x|x_1) = \delta(x - x_1)$. The conditional path can recover the target

unconditional path: $p_t(x_t) = \int_{\mathcal{M}} p_t(x_t|x_1)p_1(x_1)dx_1$. The conditional vector field $u_t(x|x_1) \in \mathcal{T}_x\mathcal{M}$ generates the conditional probability path $p_t(x|x_1)$, the Riemannian Conditional Flow Matching objective is then:

$$\mathcal{L}_{CFM} = \mathbb{E}_{t,p_1(x_1),p_t(x|x_1)} [\|v_t(x) - u_t(x|x_1)\|_g^2], \quad (2)$$

where $t \sim \mathcal{U}([0, 1])$, $x_1 \sim p_1(x_1)$, $x \sim p_t(x|x_1)$ and $\|\cdot\|_g^2$ is the norm induced by the Riemannian metric g .

3 METHODOLOGY

As shown in Figure 2, SLFM comprises three components: the **Seed Selector** $\mathbb{S}$, which scores nodes in the 1-ego net (the seed node along with its neighbors within 1-hop) and updates the seed by picking the top-scoring node. The **Hyperbolic Flow Transporter** $\mathbb{T}$, which takes a Riemannian normal distribution p_0 centered at the seed's neighborhood and learns a time-dependent vector field to transport p_0 toward an empirical target p_1 supported on labeled communities nodes. The **Community Expander** $\mathbb{E}$, which ranks candidate nodes by a hyperbolic kernel-coverage gain. This gain is computed from the geodesic distances to the endpoints generated by $\mathbb{T}$. $\mathbb{E}$ then greedily appends the node with the largest marginal gain until a stopping criterion is met. Sections 3.1–3.4 detail each component, and Algorithm 1 summarizes the overall procedure.

3.1 Hyperbolic Embedding of Nodes

We introduce notation for the hyperboloid model of hyperbolic space. Let $\langle \cdot, \cdot \rangle_L : \mathbb{R}^{d+1} \times \mathbb{R}^{d+1} \rightarrow \mathbb{R}$ denote the Minkowski inner product in the Lorentz model, $\langle \mathbf{x}, \mathbf{y} \rangle_L = -x_0y_0 + x_1y_1 + \dots + x_dy_d$. We define $\mathbb{H}^{d,K}$ to be the d -dimensional hyperboloid of constant negative curvature $-1/K$ ($K > 0$):

$$\mathbb{H}^{d,K} = \{ \mathbf{x} \in \mathbb{R}^{d+1} : \langle \mathbf{x}, \mathbf{x} \rangle_L = -K, x_0 > 0 \}. \quad (3)$$

Given a graph G , we obtain node embeddings in hyperbolic space using a Lorentz-model Hyperbolic-GCN encoder [5]. Specifically, the encoder maps each node $u \in \mathcal{V}$ to a representation $\mathbf{z}(u)$ that lies on the hyperboloid $\mathbb{H}^{d,K}$:

$$\mathbf{z}(u) = f_\theta(u; G, \mathbf{X}) \in \mathbb{H}^{d,K}. \quad (4)$$

Hyperbolic embeddings tend to cluster nodes from the same community into contiguous angular sectors, where angular proximity reflects node similarity, and radial distance encodes node centrality [12, 18]. Embedding nodes in continuous hyperbolic space aligns with our model design, allowing us to exploit geometric priors (angular and radial coordinates) for seed selection.

3.2 Seed Selector

The goal of the seed selector $\mathbb{S}$ is to assess the seed u_0 and to replace it with the highest-scoring node within the 1-ego net. It relies on three metrics: *Angular Consistency*, *Radial Conformity*, and *Structural Curvature*, which summarize directional alignment, radial depth, and local topological cohesion.

3.2.1 Angular Consistency. In hyperbolic space, nodes that belong to the same community typically concentrate within common angular sectors, where the angular coordinate is strongly correlated with node similarity [12, 18]. To capture this property, we derive the node's direction from its hyperbolic embedding and define angular

consistency to quantify the directional alignment between a node u and its 1-hop neighborhood $\mathcal{N}(u)$.

Let $o = (\sqrt{K}, 0, \dots, 0)$ denote the origin point on $\mathbb{H}^{d,K}$. We first compute the unit direction vector of u in the tangent space at o :

$$\hat{\mathbf{t}}(u) = \frac{\text{Log}_o^K(\mathbf{z}(u))}{\|\text{Log}_o^K(\mathbf{z}(u))\|_{g_o}}, \quad (5)$$

where $\text{Log}_o^K(\cdot)$ is the logarithmic map on a hyperbolic space of curvature $-1/K$ that maps embeddings to the tangent space T_o , and $\|\cdot\|_{g_o}$ denotes the Riemannian norm on the tangent space T_o induced by the Lorentzian metric.

The angular consistency is then defined as the mean alignment between the direction of u and those of its 1-hop neighbors:

$$\mathbb{S}_{\text{ang}}(u) = \frac{1}{2} \left(1 + \frac{1}{\deg(u)} \sum_{v \in \mathcal{N}(u)} \langle \hat{\mathbf{t}}(u), \hat{\mathbf{t}}(v) \rangle_{g_o} \right), \quad (6)$$

where $\deg(u)$ denotes the degree of u , and $\langle \cdot, \cdot \rangle_{g_o}$ is the Riemannian inner product on T_o induced by the Lorentzian metric. By construction, $\mathbb{S}_{\text{ang}}(u) \in (0, 1)$, with larger values corresponding to greater mean alignment between the direction of u and those of its 1-hop neighbors.

3.2.2 Radial Conformity. In hyperbolic space, nodes that belong to the centrals of the community typically occupy similar radial depths. To quantify this effect for a node u , we measure its hyperbolic radius relative to the origin o and compare it with the median radius of $\mathcal{N}(u)$. The node's radius is

$$r(u) = d_K(\mathbf{z}(u), o) = \sqrt{K} \operatorname{arccosh} \left(-\frac{\langle \mathbf{z}(u), o \rangle_L}{K} \right), \quad (7)$$

where $d_K(\cdot, \cdot)$ denotes the hyperbolic geodesic distance on $\mathbb{H}^{d,K}$. Let $\tilde{r}(u)$ denote the neighborhood's median radius, defined as the median of neighbor radii:

$$\tilde{r}(u) = \operatorname{median}\{r(v) | v \in \mathcal{N}(u)\}. \quad (8)$$

We then define radial conformity as a sigmoid of the negative absolute deviation from this median:

$$\mathbb{S}_{\text{rad}}(u) = \sigma(-\eta|r(u) - \tilde{r}(u)|), \quad (9)$$

where $\sigma(\cdot)$ denotes the sigmoid function and η is set using the neighborhood median absolute deviation as a robust scale estimator [19]. By construction, $\mathbb{S}_{\text{rad}}(u) \in (0, 1)$, with larger values corresponding to smaller deviations from the neighborhood's radial level.

3.2.3 Structural Curvature. While angular consistency and radial conformity are defined in hyperbolic space, they do not capture the node's structural role in the graph. This omission can lead the seed selector to over-score nodes with undesirable local topology, such as boundary bridges or hubs dominated by cross-community edges. We therefore introduce the unweighted augmented Forman–Ricci curvature [26, 33, 36] to characterize the node's structural role in the graph.

For each edge $e = (u, v)$, the edge curvature is defined as:

$$F_{\text{cur}}(u, v) = 4 - \deg(u) - \deg(v) + 3T_{uv}, \quad (10)$$

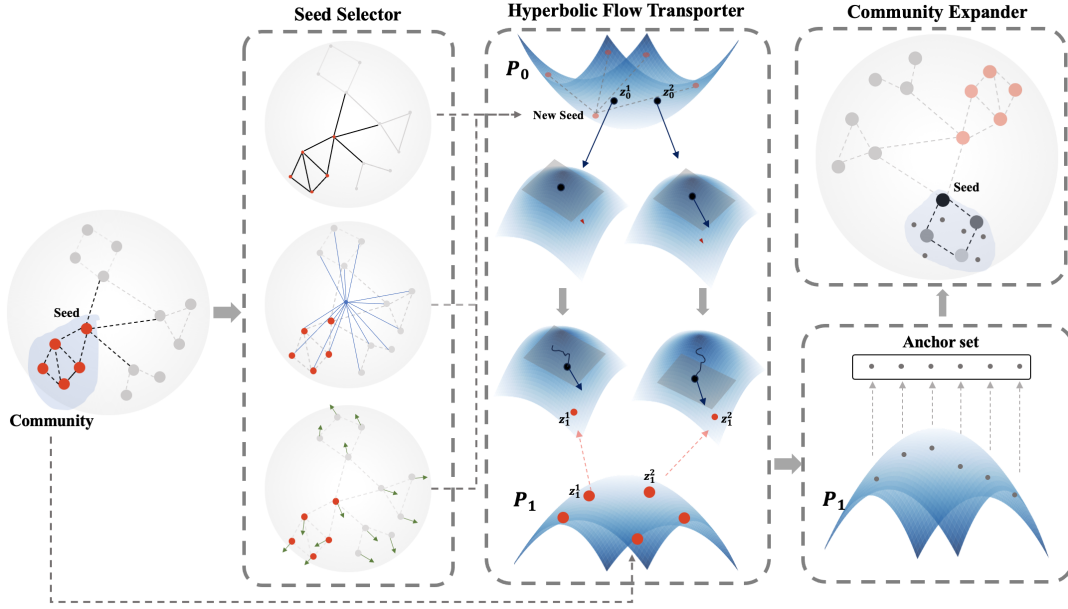


Figure 2: An illustration of the SLFM architecture. The blue region represents hyperbolic space, where community nodes are mapped into a sector oriented in the same direction. SLFM consists of three major components: Seed Selector, Hyperbolic Flow Transporter, and Community Expander. The Seed Selector scores nodes and updates the seed to the top-scoring node. The Hyperbolic Flow Transporter learns a time-dependent vector field in hyperbolic space and generates an anchor set. The Community Expander, which is based on surrogate supervision, aims to find the community the seed belongs to. For the community expander, it greedily grows from flow anchors by adding the frontier node that maximizes anchor-affinity gain at the Karcher center, yielding geometry-consistent expansion that respects boundaries and mitigates seed sensitivity.

where $T_{uv} = |\mathcal{N}(u) \cap \mathcal{N}(v)|$ is the number of triangle structures incident to e . We aggregate the edge curvatures with degree normalization to obtain the structural curvature:

$$\mathbb{S}_{\text{str}}(u) = \sigma \left(\tau \cdot \frac{1}{\deg(u)} \sum_{v \in \mathcal{N}(u)} F_{\text{cur}}(u, v) \right) \in (0, 1). \quad (11)$$

Here, $\tau > 0$ is a scaling hyperparameter (default $\tau = 0.1$) used to stabilize the value range. Higher scores indicate triangle-rich, cohesive neighborhoods, whereas lower scores suggest boundary nodes or structural hubs.

3.2.4 Score Aggregation and Seed Replacement. For a node u , we fuse angular consistency, radial conformity, and structural curvature into a single score:

$$\mathbb{S}(u) = \alpha \mathbb{S}_{\text{ang}}(u) + \beta \mathbb{S}_{\text{rad}}(u) + (1 - \alpha - \beta) \mathbb{S}_{\text{str}}(u), \quad (12)$$

where α and β are nonnegative hyperparameters controlling the relative contributions of the angular and radial terms, respectively. We enforce $\alpha, \beta \in [0, 1]$ and $\alpha + \beta \leq 1$ (default $\alpha = \beta = 0.2$).

Given a seed node u_0 , let $\mathcal{N}[u_0] = \{u_0\} \cup \mathcal{N}(u_0)$ denote the 1-ego net of u_0 . The new seed is chosen as:

$$u'_0 = \arg \max_{u \in \mathcal{N}[u_0]} \mathbb{S}(u), \quad (13)$$

and we set $u_0 \leftarrow u'_0$.

In practice, during training we apply the seed selector $\mathbb{S}$ once to the 1-ego net $\mathcal{N}[u_0]$ of the provided seed to possibly replace u_0 with a higher-quality seed $\hat{u}_0$ before constructing the source

$p_{u_0}^0$ and forming training pairs. Inference, performing the same 1-hop refinement typically yields stronger performance; however, it slightly reduces the likelihood that the original seed u_0 is included in the returned community, as the expansion is then centered around the refined seed rather than u_0 .

3.3 Hyperbolic Flow Transporter

Motivated by Flow Matching [7, 9, 21] and Riemannian bridge matching [39], we formulate seed-driven community expansion as continuous transport of probability distributions on the hyperbolic manifold $\mathbb{H}^{d,K}$. To our knowledge, this is the first application of flow matching to semi-supervised local community detection.

Let $t \in [0, 1]$ denote time. During training, the Hyperbolic Flow Transporter $\mathbb{T}$ learns a time-dependent tangent vector field $\mathbf{f}_\theta : \mathbb{H}^{d,K} \times [0, 1] \rightarrow T\mathbb{H}^{d,K}$ with $\mathbf{f}_\theta(\mathbf{z}, t) \in T_{\mathbf{z}}\mathbb{H}^{d,K}$, and transports a seed-specific source $p_{u_0}^0$ to an empirical target $p_{u_0}^1$ supported on nodes in the seed's community C_{u_0} . At inference, given a seed node u_0 , we construct $p_{u_0}^0$ from its 1-ego net $\mathcal{N}[u_0]$, draw samples $\mathbf{Z}_0 = \{\mathbf{z}_0^{(i)}\}_{i=1}^n \sim p_{u_0}^0$, and transport them under the flow induced by the learned vector field $\mathbf{f}_\theta$ to obtain endpoints $\mathbf{Y} = \{\mathbf{y}^{(i)}\}_{i=1}^n$. These endpoints serve as an anchor set for the Community Expander $\mathbb{E}$.

This formulation directly models the distributional relationship between a seed and its ground-truth community, yielding an interpolation between the seed-specific source and the community-supported target.

3.3.1 Design of Source Distribution $p_{u_0}^0$. Rather than constructing the source distribution $p_{u_0}^0$ at the single seed embedding $\mathbf{z}(u_0)$, which can be overly sensitive to noise at the seed location, we construct $p_{u_0}^0$ by aggregating nodes in the 1-ego net $\mathcal{N}[u_0]$. Specifically, we weight nodes in $\mathcal{N}[u_0]$ by the scores produced by the seed selector $\mathbb{S}$, thereby reducing sensitivity to single-point initialization and reflecting the seed's local neighborhood.

For the seed node u_0 , we approximate $p_{u_0}^0$ as a unimodal Riemannian normal distribution on the hyperbolic manifold $\mathbb{H}^{d,K}$ with density:

$$p_{u_0}^0(\mathbf{z}) \propto \exp\left(-\frac{d_K(\mathbf{z}, \mu_{u_0})^2}{2\sigma_{u_0}^2}\right). \quad (14)$$

Under this formulation, $p_{u_0}^0$ is parameterized by its Riemannian mean μ_{u_0} and variance $\sigma_{u_0}^2$.

We estimate μ_{u_0} via j iterations of the weighted Karcher mean update [17]:

$$\mu^{(j)} = \text{Exp}_{\mu^{(j-1)}}^K \left(\sum_{v \in \mathcal{N}[u_0]} \tilde{w}_v \text{Log}_{\mu^{(j-1)}}^K(\mathbf{z}(v)) \right), \quad (15)$$

where $\tilde{w}_v = \frac{w_v}{\sum_{q \in \mathcal{N}[u_0]} w_q}$ are normalized weights with raw scores

$w_v = \mathbb{S}(v)$, and the iteration is initialized at $\mu^{(0)} = \mathbf{z}(u_0)$. Unless otherwise stated, we set $j = 3$ and take $\mu_{u_0} = \mu^{(3)}$. The variance is defined from the weighted dispersion around μ_{u_0} :

$$\sigma_{u_0}^2 = \sum_{v \in \mathcal{N}[u_0]} \tilde{w}_v d_K(\mu_{u_0}, \mathbf{z}(v))^2. \quad (16)$$

This choice provides a small-variance approximation to the hyperbolic Riemannian normal distribution and yields a numerically stable source construction [29, 30]. Finally, we generate the source set $\mathbf{Z}_0 = \{\mathbf{z}_0^{(i)}\}_{i=1}^n$ by independently sampling n points from $p_{u_0}^0$.

3.3.2 Design of Target Distribution $p_{u_0}^1$. Previous work on flow matching [2, 9, 21] assumes access to i.i.d. samples from the target distribution. In our semi-supervised setting, such samples are provided by the hyperbolic embeddings of nodes in C_{u_0} . Accordingly, we define the target as the empirical measure:

$$p_{u_0}^1 := \frac{1}{|C_{u_0}|} \sum_{v \in C_{u_0}} \delta_{\mathbf{z}(v)}, \quad (17)$$

where $\delta_{\mathbf{z}(v)}$ denotes the Dirac measure at $\mathbf{z}(v)$. This target is used only during training, since C_{u_0} is unknown at inference time.

To form training pairs, we sample $v_i \sim \text{Unif}(C_{u_0})$ with replacement and set $\mathbf{z}_1^{(i)} = \mathbf{z}(v_i)$; independently, we draw $\mathbf{z}_0^{(i)} \sim p_{u_0}^0$. Pairing by index yields the dataset $\mathcal{D}_{u_0} = \{(\mathbf{z}_0^{(i)}, \mathbf{z}_1^{(i)})\}_{i=1}^n$.

3.3.3 Hyperbolic Flow Matching. Given a set of training pairs $\mathcal{D}_{u_0}$, the goal of the Hyperbolic Flow Transporter $\mathbb{T}$ is to learn a time-dependent tangent vector field on $\mathbb{H}^{d,K}$:

$$\mathbf{f}_\theta : \mathbb{H}^{d,K} \times [0, 1] \rightarrow T\mathbb{H}^{d,K}, \quad \mathbf{f}_\theta(\mathbf{z}, t) \in T_{\mathbf{z}}\mathbb{H}^{d,K}, \quad (18)$$

which transports the distribution from $p_{u_0}^0$ to $p_{u_0}^1$, where $T_{\mathbf{z}}\mathbb{H}^{d,K}$ denotes the tangent space of $\mathbb{H}^{d,K}$ at $\mathbf{z}$. Specifically, for a training

pair $(\mathbf{z}_0, \mathbf{z}_1) \in \mathcal{D}_{u_0}$, we define the geodesic interpolation as:

$$\mathbf{z}_t = \gamma_{\mathbf{z}_0, \mathbf{z}_1}(t) = \text{Exp}_{\mathbf{z}_0}^K(t \text{Log}_{\mathbf{z}_0}^K(\mathbf{z}_1)), \quad t \sim \text{Unif}[0, 1]. \quad (19)$$

In the Lorentz model, the supervised velocity is obtained by parallel-transporting the initial log-map direction along the same geodesic:

$$\mathbf{s}_t := \text{PT}_{\mathbf{z}_0 \rightarrow \mathbf{z}_t}^{\gamma_{\mathbf{z}_0, \mathbf{z}_1}}(\text{Log}_{\mathbf{z}_0}^K(\mathbf{z}_1)) \in T_{\mathbf{z}_t}\mathbb{H}^{d,K}, \quad (20)$$

where the parallel transport operator admits a closed form

$$\text{PT}_{\mathbf{z}_0 \rightarrow \mathbf{z}_t}^{\gamma_{\mathbf{z}_0, \mathbf{z}_1}}(\cdot) = \cdot + \frac{\langle \mathbf{z}_t, \cdot \rangle_L}{K - \langle \mathbf{z}_0, \mathbf{z}_t \rangle_L} (\mathbf{z}_0 + \mathbf{z}_t). \quad (21)$$

This closed-form transport is an isometry between tangent spaces along $\gamma_{\mathbf{z}_0, \mathbf{z}_1}$, ensuring $\langle \mathbf{s}_t, \mathbf{z}_t \rangle_L = 0$. The model outputs a raw vector $\hat{\mathbf{f}}_\theta(\mathbf{z}_t, t)$ which is projected onto the tangent space to ensure geometric validity:

$$\hat{\mathbf{f}}_\theta(\mathbf{z}_t, t) = \Pi_{\mathbf{z}_t}(\tilde{\mathbf{f}}_\theta(\mathbf{z}_t, t)) \in T_{\mathbf{z}_t}\mathbb{H}^{d,K}, \quad (22)$$

where $\Pi_{\mathbf{z}}$ denotes the Lorentz-orthogonal projection onto $T_{\mathbf{z}}\mathbb{H}^{d,K}$. In the Lorentz model, an explicit form is $\Pi_{\mathbf{z}}(\cdot) = \cdot - \frac{\langle \cdot, \mathbf{z} \rangle_L}{\langle \mathbf{z}, \mathbf{z} \rangle_L} \mathbf{z}$.

We train by aligning the tangent vector field with $\mathbf{s}_t$ via a Riemannian least-squares objective. Intuitively, this loss measures the mean-squared discrepancy, under the local metric $g_{\mathbf{z}_t}$, between the model's tangent velocity and the parallel-transported supervision along the geodesic, averaged over training pairs and uniformly sampled times:

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{(\mathbf{z}_0, \mathbf{z}_1) \sim \mathcal{D}_{u_0}} \mathbb{E}_{t \sim \text{Unif}[0, 1]} \left[\|\hat{\mathbf{f}}_\theta(\mathbf{z}_t, t) - \mathbf{s}_t\|_{g_{\mathbf{z}_t}}^2 \right]. \quad (23)$$

where $\|\cdot\|_{g_{\mathbf{z}_t}}$ is the Riemannian norm at $\mathbf{z}_t$, given by the Minkowski inner product restricted to $T_{\mathbf{z}_t}$ in the Lorentz model.

3.3.4 Graph-Structured Regularization. To incorporate graph structure into the learning objective of the time-dependent vector field, we augment pairwise supervision with a structure-aware target. Rather than supervising solely with the endpoint $\mathbf{z}_1 = \mathbf{z}(v)$, we learn a local mean tangent direction that summarizes the neighborhood of v . This encourages $\mathbf{f}_\theta(\mathbf{z}_t, t)$ to align with local community structure and reduces the variance of pairwise supervision.

Specifically, for each training pair $(\mathbf{z}_0, \mathbf{z}_1)$ with $\mathbf{z}_1 = \mathbf{z}(v)$, we define a local mean tangent vector by aggregating the node's neighbors:

$$\tilde{\mathbf{v}} = \sum_{q \in \mathcal{N}(v)} \tilde{w}_q \text{Log}_{\mathbf{z}_0}^K(\mathbf{z}(q)), \quad (24)$$

where the weights are initialized and normalized per Eq. 15. Intuitively, $\tilde{\mathbf{v}}$ summarizes the predominant tangent direction of v 's neighborhood in the reference space $T_{\mathbf{z}_0}\mathbb{H}^{d,K}$, reducing noise from any single endpoint. The corresponding local-mean target velocity is obtained via parallel transport:

$$\mathbf{s}_t^{(v)} := \text{PT}_{\mathbf{z}_0 \rightarrow \mathbf{z}_t}^{\gamma_{\mathbf{z}_0, \mathbf{z}_1}}(\tilde{\mathbf{v}}). \quad (25)$$

This transport carries the neighborhood summary along the geodesic to $T_{\mathbf{z}_t}\mathbb{H}^{d,K}$ while preserving Lorentzian inner products, ensuring geometric consistency of the supervision.

The graph-structured regularization loss is

$$\mathcal{L}_{\text{graph}} = \mathbb{E}_{(\mathbf{z}_0, \mathbf{z}_1) \sim \mathcal{D}_{u_0}} \mathbb{E}_{t \sim \text{Unif}[0, 1]} \left[\|\hat{\mathbf{f}}_\theta(\mathbf{z}_t, t) - \mathbf{s}_t^{(v)}\|_{g_{\mathbf{z}_t}}^2 \right]. \quad (26)$$

For the Hyperbolic Flow Transporter $\mathbb{T}$, the overall objective combines flow matching with the graph-structured term:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{FM}} + \lambda_{\text{graph}} \mathcal{L}_{\text{graph}}, \quad (27)$$

where λ_{graph} is a hyperparameter.

3.4 Community Expander

Given a seed u_0 , we draw n samples $\mathbf{Z}_0 = \{\mathbf{z}_0^{(i)}\}_{i=1}^n$ independently from $p_{u_0}^0$ (cf. Section 3.3.1). Each sample is then transported by integrating the projected vector field $\hat{\mathbf{f}}_\theta(\mathbf{z}_t, t)$ over $t \in [0, 1]$ (cf. Section 3.3.3). The resulting endpoints, referred to as anchors, are denoted by $\mathbf{y}^{(i)} := \mathbf{z}_1^{(i)}$, and we write $\mathbf{Y} = \{\mathbf{y}^{(i)}\}_{i=1}^n \subset \mathbb{H}^{d,K}$.

Let ν_Y be the unweighted Karcher mean of $\mathbf{Y}$, computed via fixed-point updates

$$\nu_Y^{(j)} = \text{Exp}_{\nu_Y^{(j-1)}}^K \left(\frac{1}{n} \sum_{i=1}^n \text{Log}_{\nu_Y^{(j-1)}}^K(\mathbf{y}^{(i)}) \right), \quad (28)$$

initialized at $\nu_Y^{(0)} = \mathbf{y}^{(1)}$ and run for $j \in \{1, 2, 3\}$, after which we set $\nu_Y := \nu_Y^{(3)}$. We adopt an adaptive bandwidth

$$h := \text{median}_{i=1, \dots, n} d_K(\nu_Y, \mathbf{y}^{(i)}), \quad (29)$$

We then define a Gaussian kernel on the hyperbolic distance [8]

$$k_h(\mathbf{p}, \mathbf{y}) = \exp\left(-\frac{d_K(\mathbf{p}, \mathbf{y})^2}{2h^2}\right), \quad (30)$$

and the average affinity to the anchors

$$\bar{k}_h(\mathbf{p}; \mathbf{Y}) = \frac{1}{|\mathbf{Y}|} \sum_{\mathbf{y} \in \mathbf{Y}} k_h(\mathbf{p}, \mathbf{y}). \quad (31)$$

For a node u , we write $\bar{k}_h(u; \mathbf{Y}) = \bar{k}_h(\mathbf{z}(u); \mathbf{Y})$. With the median-based bandwidth h , the Gaussian kernel turns geodesic proximity into a smooth affinity, and $\bar{k}_h(u; \mathbf{Y})$ acts as a soft coverage score—high when u lies near many anchors and low otherwise.

For a nonempty node set $C \subseteq \mathcal{V}$, let its Karcher center [1] be

$$\mu_C = \arg \min_{\mathbf{q} \in \mathbb{H}^{d,K}} \frac{1}{|C|} \sum_{v \in C} d_K(\mathbf{q}, \mathbf{z}(v))^2, \quad (32)$$

approximated by three fixed-point steps initialized at $\mathbf{z}(u_0)$. We score a community by the anchor affinity of its center:

$$\phi(C) := \bar{k}_h(\mu_C; \mathbf{Y}). \quad (33)$$

Given C , define the 1-hop frontier

$$\mathcal{A}(C) := \left(\bigcup_{v \in C} \mathcal{N}(v) \right) \setminus C. \quad (34)$$

At each step, we consider the marginal gain of adding a frontier node u :

$$\Delta(u \mid C) := \phi(C \cup \{u\}) - \phi(C). \quad u \in \mathcal{A}(C) \quad (35)$$

Let $u^* \in \arg \max_{u \in \mathcal{A}(C)} \Delta(u \mid C)$. If $\Delta(u^* \mid C) > 0$, we add u^* to C and update both μ_C and $\mathcal{A}(C)$; otherwise, we delete u^* from the frontier and continue the search. The procedure repeats until $\mathcal{A}(C) = \emptyset$. In words, each iteration adds the frontier node that maximally increases affinities of the community's center with the anchor set, and stops once no candidate yields a positive gain.

4 EXPERIMENTS

4.1 Experimental Settings

We utilize four real-world datasets, including Amazon [24], DBLP [41], LiveJournal [3], YouTube [25]. In our experiments, the datasets are preprocessed following Node2Vec [14], HGCN [5], and SLRL [28].

SLFM uses a graph neural network (e.g., HGCN) to obtain node embeddings in hyperbolic space, which requires node-level features. However, several of the datasets we use do not provide node attributes. To fairly reflect SLFM's performance across datasets with heterogeneous attribute availability, we adopt the following unified setup: we derive Euclidean node features for all datasets using Node2Vec on the raw graph and use these as fixed inputs to HGCN. We then train HGCN in the hyperboloid model to produce hyperbolic embeddings, which are fed to SLFM. Unless stated otherwise, the Node2Vec features are frozen and not jointly fine-tuned with HGCN. Importantly, the node embeddings for Node2Vec and HGCN were learned in an unsupervised manner, thereby avoiding any risk of supervision leakage. For each network, we use 700 communities for training, 70 for validation, and the remainder for testing.

4.2 Evaluation Metrics

To evaluate the performance of the algorithm, we employ Precision and Recall [10], F1-score [31], and Jaccard [15] to quantify the agreement between the detected communities and the ground-truth communities. For the semi-supervised global community detection [4, 16, 40, 43] setting, we take as the detected community the one that contains the given node among all communities produced by the algorithm. Higher values of these metrics indicate better community quality. All metrics are macro-averaged over test seeds; F1 and Jaccard are computed per seed and then averaged.

4.3 Comparison Methods

We compare SLFM with one local community detection method M [23]; two semi-supervised global community detection methods SEAL [43] and CLARE [40]; three semi-supervised local methods, SLSS [27], CommunityAF [6] and SLRL [28]. The number of communities generated by CLARE and SEAL is set to 5000 on Amazon, DBLP, LiveJournal, and YouTube. Groups of ground-truth communities, comprising 200 communities, are selected as the known communities for the experiments. SEAL, CLARE, and SLFM are run twice for each group. The average metrics from these two runs are used as the metric for each group. The average metrics across the two groups are taken as the final metrics for the algorithm.

4.4 Experimental Results

Comparison With Local Community Detection Methods. Here, we compare both unsupervised and semi-supervised local community detection methods, including the M method, SLSS, CommunityAF, SLRL, and SLFM on four real-world datasets. Table 1 presents the results of these methods.

Table 1 shows the performance of SLFM is significantly better than that of M method, SLSS, CommunityAF, and SLRL. Specifically, on DBLP, YouTube, and LiveJournal datasets, SLFM outperforms M method, SLSS, CommunityAF, and SLRL. SLFM is worse than other methods for Amazon. In all datasets, SLSS, CommunityAF, and SLRL

Table 1: Results of M, SLSS, CommunityAF, SLRL, and SLFM. ComAF is the abbreviation of CommunityAF. The best results are bolded, and the second-best results are underlined.

Dataset	Amazon				DBLP				LiveJournal				YouTube			
Metric	P	R	F	J	P	R	F	J	P	R	F	J	P	R	F	J
M	0.798	0.784	0.751	0.649	0.591	0.588	0.546	0.433	0.561	0.718	0.601	0.508	0.275	0.296	0.236	0.152
SLSS	<u>0.838</u>	0.860	0.820	0.729	0.609	0.712	0.618	0.499	0.573	<u>0.773</u>	<u>0.635</u>	<u>0.523</u>	0.239	<u>0.362</u>	0.233	0.146
ComAF	0.821	0.896	0.836	0.742	0.507	0.782	0.598	0.451	<u>0.657</u>	0.630	0.592	0.477	0.344	0.287	0.269	0.172
SLRL	0.846	0.927	<u>0.870</u>	0.790	<u>0.664</u>	0.718	<u>0.654</u>	<u>0.531</u>	<u>0.657</u>	0.694	0.572	0.460	<u>0.531</u>	0.264	<u>0.284</u>	<u>0.184</u>
SLFM	0.835	<u>0.921</u>	0.872	<u>0.773</u>	0.683	<u>0.762</u>	0.682	0.540	0.686	0.775	0.651	0.551	0.543	0.401	0.319	0.219
Improve	↓1.30%	↓0.65%	↑0.23%	↓2.15%	↑2.86%	↓2.56%	↑4.28%	↑1.69%	↑4.41%	↑0.26%	↑2.52%	↑5.35%	↑2.26%	↑10.77%	↑12.32%	↑15.54%

Table 2: Results of SEAL, CLARE, and SLFM. The best results are bolded, and the second-best results are underlined.

Dataset	Amazon				DBLP				LiveJournal				YouTube			
Metric	P	R	F	J	P	R	F	J	P	R	F	J	P	R	F	J
SEAL	0.835	0.879	0.836	0.750	0.643	<u>0.671</u>	0.631	0.507	0.759	0.681	0.679	0.550	<u>0.442</u>	0.363	<u>0.355</u>	0.248
CLARE	0.827	0.792	0.791	0.692	0.594	0.640	0.593	0.480	0.695	0.652	0.641	0.525	0.396	0.350	0.324	0.220
SLFM	0.852	0.910	0.861	0.788	0.687	0.742	0.675	0.549	<u>0.731</u>	0.752	0.685	0.573	0.503	0.367	0.376	0.259
Improve	↑1.67%	↑3.06%	↑2.62%	↑4.64%	↑7.34%	↑9.92%	↑8.00%	↑10.46%	↓3.69%	↑10.42%	↑0.88%	↑3.62%	↑12.78%	↑2.23%	↑5.62%	↑4.02%

are better than M method. The reason why SLRL, CommunityAF, SLSS, and SLFM are superior to M method on most datasets is that they use known communities to guide community detection. The M method shows good performance on the LiveJournal dataset, which indicates that the community structure on the LiveJournal dataset is consistent with the assumptions of the community structure of M method.

Overall, SLFM shows the strongest effectiveness, with higher F1 and Jaccard in aggregate and with competitive precision and recall. The advantage becomes more salient when communities are not small. Larger communities increase the separation between the source and target distributions in hyperbolic space, which provides more informative velocity supervision for flow matching and yields clearer improvements. When communities are very small, essentially limited to the seed and its 1-hop neighborhood, the source and target nearly coincide and the supervision signal weakens, so the performance of SLFM narrows. Baseline methods can exceed SLFM on isolated metrics in specific cases, but this does not change the overall trend.

Comparison With Semi-Supervised Community Detection Methods. To comprehensively evaluate the effectiveness of our proposed method, we conducted extensive experiments on four real-world datasets, with the results summarized in Table 2. The results show that SLFM consistently achieves either the best or second-best performance across all datasets and significantly outperforms the state-of-the-art semi-supervised community detection methods SEAL and CLARE. In particular, SLFM achieves the highest scores across all evaluation metrics on Amazon and DBLP datasets, with Recall improvements of 3.06% and 9.92% and Jaccard gains of 4.64% and 10.46%, respectively, demonstrating its superior capability in detecting communities within academic networks. On LiveJournal dataset, although Precision is slightly lower than that of SEAL, SLFM achieves superior performance in Recall, F1-score, and Jaccard, with Recall improving by 10.42%, indicating that SLFM remains highly effective in large-scale social networks. Even on the

noisy and structurally complex YouTube dataset, SLFM maintains its superiority, with improvements of 12.78% in Precision and 5.62% in F1-score, demonstrating its robustness and strong generalization ability in challenging network environments. Overall, SLFM consistently delivers state-of-the-art performance and demonstrates excellent scalability across diverse network scenarios. Its advantages become particularly evident in large-scale and structurally heterogeneous networks, highlighting the strong potential and practical value of SLFM for real-world community detection applications.

Stress test with 100 low-quality seeds. To evaluate the robustness of SLFM to seed quality, we conducted a stress test using 100 low-quality seeds across all datasets, including leaf nodes, boundary nodes, and hub nodes. These nodes typically provide weak or ambiguous structural cues, posing significant challenges for seed-based methods.

Even under these adverse conditions, SLFM maintains stable performance with only slight drops in Precision, Recall, and F1-score, demonstrating strong robustness. This result confirms that SLFM can effectively handle poor-quality seeds and reliably detect target communities.

4.5 Ablation Study

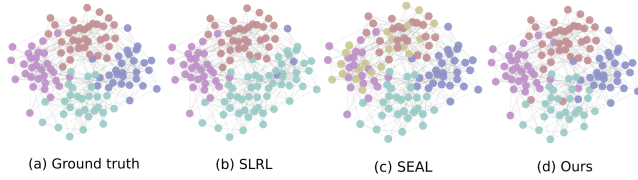
To evaluate the contribution of each core component to the overall performance of SLFM, we conducted a series of ablation experiments, and the results are presented in Table 4. As shown in the table, removing any single component leads to a noticeable degradation in both Precision and Recall on Amazon and DBLP datasets. Among them, the removal of the seed selector (w/o SS) causes the most significant performance drop, highlighting the critical importance of high-quality seed selection during inference. Similarly, eliminating the structural curvature term (w/o S_{str}) also results in a substantial decline in performance, indicating that structural information plays a crucial role in accurate community detection. Although the removal of the graph-structured regularization term

Table 3: The Stress test with 100 low-quality seeds. Results of SLSS, SLRL and SLFM. The best results are bolded.

Dataset	Amazon				DBLP				LiveJournal				YouTube			
Metric	P	R	F	J	P	R	F	J	P	R	F	J	P	R	F	J
SLSS	0.725	0.743	0.643	0.639	0.556	0.585	0.547	0.419	0.550	0.573	0.576	0.476	0.215	0.307	0.208	0.135
SLRL	0.711	0.692	0.694	0.577	0.520	0.557	0.509	0.399	0.607	0.562	0.540	0.436	0.345	0.254	0.271	0.179
SLFM	0.814	0.886	0.852	0.768	0.675	0.720	0.666	0.537	0.671	0.738	0.648	0.546	0.499	0.358	0.309	0.216

Table 4: Ablations on SLFM: core components. w/o X means model without X. w/o SS means model without the seed selector at inference.

Setting	Amazon				DBLP			
	P	R	F	J	P	R	F	J
w/o $\$_{ang}$	0.819	0.899	0.856	0.755	0.671	0.743	0.666	0.529
w/o $\$_{rad}$	0.820	0.895	0.854	0.753	0.673	0.744	0.669	0.527
w/o $\$_{str}$	0.802	0.877	0.837	0.738	0.657	0.724	0.651	0.517
w/o SS	0.772	0.838	0.810	0.703	0.631	0.694	0.630	0.493
w/o λ_{graph}	0.831	0.916	0.868	0.769	0.680	0.758	0.679	0.537
SLFM	0.835	0.921	0.872	0.773	0.683	0.762	0.682	0.540

**Figure 3: Community detection result visualization on LFR benchmark network.**

(w/o λ_{graph}) causes a relatively smaller decrease in performance, the results demonstrate that it provides valuable complementary benefits for representation learning.

Overall, these results confirm that each component of SLFM makes a positive contribution to the final model performance. In particular, the seed selector and structural information are essential for improving the robustness of community detection, while the angular and radial consistency terms, together with the graph-structured regularization, further enhance the model’s ability to capture the underlying community structure.

4.6 Case Study

The visualization in Figure 3 demonstrates that the community detection conducted by SLFM is almost identical to the ground truth. The SLRL tends to assign most nodes to fewer communities, and SEAL tends to detect more communities than the ground truth.

5 RELATED WORK

Semi-supervised community detection has received sustained attention. Many algorithms have been developed based on different frameworks, such as generative methods [6, 43], reinforcement learning based methods [40], and subgraph similarity frameworks [4]. For example, Bespoke [4] learns structural motifs and community-size priors from a few labeled examples, ranks seeds by motif support, and then breadth-first expands each seed to a sampled target size to recover communities that match both structure

and scale. SEAL[43] frames community detection as seed expansion learned via a GAN, which is a GNN generator sequentially grows communities while a discriminator evaluates realism, with the generator trained by policy gradient to learn effective expansion heuristics. CLARE[40] uses a two-stage “locate-then-rewrite” pipeline where a Community Locator retrieves subgraphs similar to training communities and a reinforcement-learning-enhanced Community Rewriter adds or drops nodes to refine the final community.

However, in some cases, users are only interested in the community to which a given node belongs. To mine a community for a given node, researchers propose many methods. CommunityAF [6] proposed an example-based community search method that trains an autoregressive flow with structure-aware features to quickly locate candidate subgraphs and then performs limited generation/refinement to recover communities matching the exemplars. M method [23] defines community via an extended subgraph-degree/local objective and extracts a source-vertex’s community using greedy algorithms. SLSS [27] used structural similarity (via graph-kernel features) between a few known communities and the seed’s neighborhood to guide seed-conditioned expansion and recover the target community. SLRL [28] Casts local community detection as a reinforcement-learning policy over seed-centric expansions, with an extractor encoding local structure and an RL-trained expander that adds/drops nodes to maximize a reward. Compared with existing approaches, we propose a flow-matching-based semi-supervised community detection algorithm that learns a time-dependent vector field along the transport path from the seed to the community distribution, and leverages this field at inference to provide supervisory signals that guide seed expansion.

6 CONCLUSION

In this paper, we revisit semi-supervised local community detection through surrogate supervision. SLFM links three simple stages: a Seed Selector that relocates the query to a reliable anchor, a Hyperbolic Flow Transporter that learns a seed-conditioned flow toward the community, and a Community Expander that turns flow-induced affinities into the final set. The selector stabilizes the flow, and the flow guides expansion. Experiments on real-world datasets show SLFM is both effective and efficient. As future work, we will explore stronger flow-matching objectives and alternative geometric priors.

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A Notations

Notations used in this paper, along with their corresponding description, are shown in Table 5.

B Riemannian Geometry

A Riemannian manifold $(\mathcal{M}, g)$ is a smooth manifold $\mathcal{M}$ endowed with a Riemannian metric tensor g . At each point $x \in \mathcal{M}$, the tangent space $T_x\mathcal{M}$ is a vector space representing the set of all possible directions of curves passing through x . The Riemannian metric g defines an inner product on the tangent space, which allows to compute lengths of tangent vectors, angles between them. Given a point in the manifold $x \in \mathcal{M}$, the exponential map projects a vector v in the tangent space at x to the manifold $\exp_x(v) : T_x\mathcal{M} \rightarrow$

Algorithm 1 SLFM**Require:** graph $G = (\mathcal{V}, \mathcal{E}, \mathbf{X})$, training set $\mathcal{D}$

```

1: Encode  $\mathbf{X}$  to  $\mathbf{z}(\mathbf{X})$  by utilize Equation 4
2: for epoch = 1, ... do
3:   Sample seeds  $u_0 \subset \mathcal{D}$  uniformly
4:   Compute  $\mathbb{S}(u)$  of  $\mathcal{N}[u_0]$  by use Equation 12
5:   updating a new seed  $u_0$  by used Equation 13
6:   Construct  $p_{u_0}^0$  distribution by Equation 14
7:   Sample  $v \subset C_{u_0}$  as  $\mathbf{z}_{u_0}^0$  uniformly
8:   Sample  $\mathbf{z}_{u_0}^0$  from  $p_{u_0}^0$  uniformly
9:   Sample  $\mathbf{z}_{u_0}^1 \subset C_{u_0}$  uniformly
10:  Construct  $\mathbf{Z} = \{\mathbf{z}_0^i, \mathbf{z}_1^i\}_{i=1}^n$ 
11:  for  $i = 1, \dots$  do
12:    Train  $\mathbf{f}_\theta$  on  $\mathbf{Y}$  with Equation 26
13:  end for
14: end for
15: Sample seeds  $u_0 \subset \mathcal{V}$  uniformly
16: Construct  $\mathbf{Y}$  with  $u_0$  and  $\mathbf{f}_\theta$ 
17: Expander node based on  $\mathbb{E}$  return  $C_{u_0}$ 

```

Table 5: Glossary of Notations.

Notation	Description
G	The Graph
$\mathcal{V}$	Nodes in Graph G
$\mathcal{E}$	Edges in Graph G
$\mathbf{X}$	The feature matrix
$\mathbf{x}(u)$	Feature vector of node u
u_0	The given seed node
C_{u_0}	The target community to which u_0 belongs
C^i	The i -th community
$\mathbb{S}$	The seed selector
$\mathbb{T}$	Hyperbolic Flow Transporter
$\mathbb{E}$	Community Expander
$\mathbb{H}^{d,K}$	The d -dimensional hyperboloid of curvature $-1/K$
K	Curvature
$\langle \cdot, \cdot \rangle_g$	The Riemannian inner product
$\mathbf{z}(\cdot)$	The hyperbolic embedding of node
o	The origin point on hyperbolic model
$\mathcal{N}(u)$	The 1-hop neighborhood of node u
$\mathcal{N}[u]$	The 1-ego net of node u
$\deg(u)$	The degree of node u
$\ \cdot\ _{g_{z_t}}$	Riemannian norm at z_t
$k_h(\mathbf{p}, \mathbf{y})$	Gaussian kernel on the hyperbolic distance
$d_K(\cdot, \cdot)$	The hyperbolic geodesic distance
$\tilde{r}(u)$	The neighborhood median radius
$\mathbf{f}_\theta$	A time-dependent tangent vector field
$T_z \mathbb{H}^{d,K}$	Tangent space at point $\mathbf{z}$
$p_{u_0}^0$	Source distribution at single seed embedding
$p_{u_0}^1$	Target distribution
Log_o^K	Logarithmic map
Exp_o^K	Exponential map
$\text{PT}_{z_0 \rightarrow z_t}$	Parallel transport operator from z_0 to z_t

 M .

$$\exp_x(v) = \cosh(\|v\|_{\mathcal{L}})x + \sinh(\|v\|_{\mathcal{L}}) \frac{v}{\|v\|_{\mathcal{L}}} \quad (36)$$

The logarithmic map projects a point y in the manifold to the tangent space of x , $\log_x(y) : M \rightarrow T_x M$, serving as the inverse of the exponential map.

$$\log_x(y) = \frac{\text{arcosh}(-(x, y)_{\mathcal{L}})}{\sqrt{(x, y)_{\mathcal{L}}^2 - 1}} (y + (x, y)_{\mathcal{L}} x) \quad (37)$$

Both the exponential and logarithmic maps are defined locally around the point x , with their domains of definition dependent on the curvature and the injectivity radius of the manifold.

A connection ∇ on $\mathcal{M}$ defines how to differentiate vector fields along curves, enabling the notion of parallel transport. The connection ∇ is called Levi-Civita connection if it preserves the metric and is torsion-free, i.e., for any vector fields X and Y , $\nabla_X Y - \nabla_Y X = [X, Y]$, where $[X, Y]$ is the Lie bracket of the vector fields X and Y . The parallel transport carries the vector in one tangent space to another along the geodesic, which is the shortest curve connecting two points in the manifold. Given a smooth curve $\gamma(t) : [x, y] \rightarrow \mathcal{M}$, a vector field $V(t)$ along γ is parallel transported if $\nabla_{\dot{\gamma}(t)} V(t) = 0$. The parallel transport map $\text{PT}_\gamma : T_{\gamma(a)} \mathcal{M} \rightarrow T_{\gamma(b)} \mathcal{M}$ is the linear isometry that maps a vector at the start of the curve to its parallel-transported version at the end.

In Riemannian geometry, the curvature is the notion to measure how a smooth manifold deviates from being flat. Suppose that $(\mathcal{M}, g)$ is an d -dimensional Riemannian manifold, equipped with Levi-Civita connection ∇ . The Riemannian curvature of $\mathcal{M}$ is a map that takes smooth vector fields X, Y and Z , and returns the vector field:

$$\mathbf{R}(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z. \quad (38)$$

if $\mathbf{R} \equiv 0$ everywhere, the manifold is flat, and parallel transport depends only on the endpoints, not the path. The Ricci curvature Ric of a Riemannian manifold is a symmetric rank-2 tensor obtained by contraction from the Riemannian curvature: $\text{Ric}(X, Y) = \sum_{i=1}^M \mathbf{R}(e_i, X, Y, e_i)$, where $\{e_i\}$ is an orthonormal basis. On a geodesic $\gamma(t)$ with unit speed $\dot{\gamma}(t)$, the Ricci curvature governs the rate of change of the volume element along the geodesic.

The d -dimensional Lorentz model equipped with negative constant curvature $\kappa < 0$ is a Riemannian manifold denoted by $\mathbb{H}^{d, \kappa}$. The corresponding Riemannian metric tensor $\mathbf{R} = \text{Diag}(-1, 1, \dots, 1)$, is defined in $(d+1)$ -dimensional Minkowski space whose origin is $(\sqrt{-1/\kappa}, 0, \dots, 0)$, i.e., $\mathbb{H}_\kappa^d = \{\mathbf{x} \in \mathbb{R}^{d+1} | \langle \mathbf{x}, \mathbf{x} \rangle_{\mathbb{H}} = \frac{1}{\kappa}\}$, where the Lorentzian inner product is $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbb{L}} = -x_0 y_0 + \sum_{i=1}^d x_i y_i = \mathbf{x}^T \mathbf{R} \mathbf{y}$. The distance between two points is given by

$$d_{\mathbb{H}}(\mathbf{x}, \mathbf{y}) = \cosh^{-1}(-\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbb{H}}). \quad (39)$$

Given $x \in \mathbb{H}^{d, \kappa}$, the tangent space $\mathcal{T}_x \mathbb{H}^{d, \kappa} = \{\mathbf{y} \in \mathbb{R}^{d+1} | \langle \mathbf{y}, \mathbf{x} \rangle_{\mathbb{H}} = 0\}$ is the orthogonal space of $\mathbb{H}^{d, \kappa}$ at x with respect to the Lorentzian inner product.

C Algorithm

The overall Algorithm of the method SLFM is shown as Algorithm 1.

D Datasets and Baselines

D.1 Datasets

We conduct experiments on four real-world datasets, which are described in detail as follows:

- **Amazon.** Amazon is collected by crawling Amazon website. If a product i is frequently co-purchased with product j , the graph contains a directed edge from i to j . Each product category provided by Amazon defines a ground-truth community.
- **DBLP.** DBLP is a co-authorship network where two authors are connected if they publish at least one paper together. Publication venue, e.g. journal or conference, defines an individual ground-truth community; authors who published to a certain journal or conference form a community.
- **LiveJournal.** LiveJournal is a free online community and it allows users to declare which other members are their friends they belong.
- **Youtube.** Youtube is a video-sharing web site that includes a social network. In the Youtube social network, users form friendship each other and users can create groups which other users can join. And the user-defined groups as ground-truth communities.

Table 6: Statistics on real-world datasets.

Datasets	Nodes	Edges	communities
Amazon	13,178	33,767	4,517
DBLP	114,095	466,761	4,559
LiveJournal	316,606	4,945,140	4,510
Youtube	216,544	1,393,206	2,865

D.2 Baselines

We compare SLFM with one local community detection method (M), two semi-supervised global community detection methods (SEAL and CLARE), and three semi-supervised local methods (SLSS, CommunityAF and SLRL). The detailed descriptions of all baselines are as follows:

- **M** is a local community detection algorithm based on the extension of degree from vertex to sub-graph, which treats nodes as elements to be greedily added or removed for modularity optimization, and aims to find compact, high-modularity communities for a given source vertex in large networks.
- **SEAL** is a semi-supervised community detection framework based on generative adversarial networks and reinforcement

learning, which treats community generation as a sequential decision process and aims to learn heuristics from training data to discover communities that match the characteristics of the given examples.

- **CLARE** is a semi-supervised community detection algorithm based on subgraph matching and deep reinforcement learning, which treats community discovery as a two-step process of locating candidate subgraphs and then intelligently rewriting their structure, and aims to find targeted communities that closely match the characteristics of the few labeled examples provided.
- **SLSS** is a semi-supervised local community detection algorithm based on structural similarity. It addresses the Semi-Supervised Local Community Detection (SLCD) problem, which aims to find the local community containing a given node using only local information and a few known communities as prior knowledge.
- **CommunityAF** is an example-based community search algorithm built on autoregressive flow modeling, which aims to efficiently discover communities similar to given examples by capturing their latent structural and content patterns.
- **SLRL** is a semi-supervised algorithm that uses reinforcement learning to find a specific node's community. It first extracts a rough local community around the node, then uses this structure to select similar known communities for training a specialized model that expands this into the final, more accurate community.

E Reproducibility

Experiments are conducted on NVIDIA GeForce RTX 4090 GPU of 24GB memory*8, and AMD EPYC 9654 CPU 315 with 96-Core Processor.

F Broader Impact

Our work brings together two previously distinct research areas, semi-supervised local community detection in graphs and deep generative modeling on geometric manifolds. We model the expansion from a seed node to its full community as a continuous transport of probability distributions on a learned hyperbolic manifold, enabling principled, non-heuristic community generation through tools from continuous generative modeling and hyperbolic geometry. Practically, our proposed SLFM model establishes a new class of accurate and robust tools for local network exploration, outperforming methods limited by exemplar communities and seed nodes. Positive societal impacts lie in the model's enhanced accuracy and reliability for local community detection, allowing for more precise and targeted analysis of complex real-world networks from minimal supervision. None of negative societal impacts we feel must be specifically highlighted.